

Online Retail II – EDA & RFM Analysis

1. Project Objective

The dataset contains transaction history from a UK-based online retailer. The main goal of the analysis was to understand customer purchasing behavior and segment customers using RFM (Recency, Frequency and Monetary) analysis.

Business question: How can transaction history be used to identify different types of customers and support targeted customer strategies?

2. Dataset Overview

Item	Result
Initial rows	1,067,371
Initial columns	8
Columns	Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, Country
Initial missing Description	4,382
Initial missing Customer ID	243,007 (22.77%)
Exact duplicate rows	34,335
Date range	01-Dec-2009 to 09-Dec-2011

3. Initial Data Checks

Why: Before analysis, we checked data types, missing values, duplicates, date range and unusual numeric values. This prevents incorrect customer or revenue calculations.

InvoiceDate was converted to a proper datetime field. The date conversion succeeded for all rows. Quantity had large positive and negative values, and Price also contained extreme values, so these were investigated before calculating revenue.

4. Cleaning Decisions

Issue	What we found	Decision
Exact duplicates	34,335 rows	Removed
Cancelled invoices	19,494 raw rows	Removed
Negative quantity	3,457 raw rows were negative but not cancellation invoices	Investigated remaining negative/zero quantities removed
Negative prices	5 accounting-adjustment rows	Excluded by final Price > 0 filter
Missing Customer ID	Large number of rows without customer identity	Removed for customer-level RFM
Zero-price transactions	70 rows after customer/quantity filtering	Removed from RFM dataset

The negative-price rows were identified as entries such as “Adjust bad debt”, rather than normal customer purchases. Missing Customer IDs were excluded because RFM is a customer-level analysis and those transactions cannot be assigned to a customer.

Final RFM transaction dataset: 779,425 valid transaction rows and 5,878 identifiable customers.

5. Revenue Creation

Why: The dataset had Quantity and Price but no direct transaction revenue column. We created Revenue = Quantity × Price. This allowed us to measure customer monetary value.

Revenue was highly right-skewed: median transaction revenue was £12.48 while the mean was about £22.29. This is consistent with a retail dataset containing some large/wholesale purchases.

6. Basic Customer-Level EDA

We checked how many customers and invoices existed and how purchase behavior was distributed.

Metric	Result
Unique customers before zero-price removal	5,881
Unique customers in final RFM data	5,878
Unique invoices in final RFM data	36,975
Average orders/customer	6.29
Median orders/customer	3
One-time customers	1,626 (~27.6%)
Repeat customers	4,255 (~72.4%)
Average revenue/customer	£2,955.90
Median revenue/customer	£867.74

Customer revenue was also strongly right-skewed. The highest customer revenue was about £580,987, showing that a small number of customers contribute disproportionately large amounts.

7. RFM Analysis

Recency: number of days since the customer's latest purchase. Lower is better because the customer purchased more recently.

Frequency: number of unique invoices/orders placed by the customer. Higher means the customer purchased more often.

Monetary: total revenue generated by the customer. Higher means greater historical customer value.

The reference date was set to one day after the latest transaction date: 10-Dec-2011 12:50. This makes the most recent customer purchase have a recency of approximately 1 day.

Metric	Mean	Median	Min	Max
Recency (days)	201.33	96	1	739
Frequency (orders)	6.29	3	1	398
Monetary (£)	2,955.90	867.74	2.95	580,987.04

8. RFM Scoring Card

Each RFM measure was converted into a 1–5 score using quintile-based scoring. For Recency the direction was reversed: recent customers received higher scores. For Frequency and Monetary, higher values received higher scores.

Score	Recency	Frequency	Monetary
5	Most recent	Highest frequency	Highest spend
4	Relatively recent	Relatively high	Relatively high
3	Middle	Middle	Middle
2	Relatively old	Relatively low	Relatively low
1	Least recent	Lowest frequency	Lowest spend

Frequency contained many tied values, especially Frequency = 1, so a direct qcut produced duplicate bin edges. We solved this by ranking Frequency first and then applying quintiles. The original Frequency values were not changed.

An RFM code combines the three scores. For example, 545 means R=5, F=4, M=5. The total RFM score ranges from 3 to 15. The final average score was about 9.01.

9. Customer Segmentation

We created six mutually exclusive, business-oriented segments using the R, F and M scores. The rules were applied in priority order so each customer received exactly one segment.

Segment	Customers	Revenue	% Customers	% Revenue
Champions	1,297	£11,859,591.50	22.07%	68.26%
At Risk	1,067	£1,919,927.90	18.15%	11.05%
Loyal Customers	701	£1,856,667.02	11.93%	10.69%
Regular / Mid-Value	1,090	£1,022,467.71	18.54%	5.88%
Potential Customers	443	£392,267.02	7.54%	2.26%
Inactive / Low Value	1,280	£323,883.11	21.78%	1.86%

10. Segment Behavior

Segment	Avg Recency	Avg Frequency	Avg Monetary	Avg Order Value
Champions	19.96 days	17.11	£9,143.86	£534.55
At Risk	379.99 days	4.18	£1,799.37	£430.38
Loyal Customers	84.37 days	7.74	£2,648.60	£342.37
Regular / Mid-Value	75.52 days	2.51	£938.04	£373.98
Potential Customers	28.11 days	1.46	£885.48	£606.29
Inactive / Low Value	467.32 days	1.19	£253.03	£213.36

Main business finding: Champions represent about 22% of customers but generate about 68% of revenue. This shows strong revenue concentration in a relatively small customer group.

At Risk customers have much higher historical value and frequency than Inactive / Low Value customers, but their average recency is around 380 days. This makes them a distinct re-engagement population.

11. Product-Level Analysis

We also examined products purchased by each segment. Three measures were used: product revenue, quantity sold and unique customers buying the product. POSTAGE and Manual were removed only from this product-preference analysis because they are not merchandise products.

Products repeatedly appearing across segments included White Hanging Heart T-Light Holder, Regency Cakestand 3 Tier, Baking Set 9 Piece RetroSpot, Assorted Colour Bird Ornament, Party Bunting and Paper Chain Kit 50's Christmas.

We also checked revenue per customer. This showed why revenue alone can be misleading: some very large product revenues were generated by only one or a few customers. Therefore, customer reach was used alongside revenue when interpreting product preferences.

12. Final EDA Story

- The retailer has 5,878 identifiable customers in the final RFM dataset.
- Customer behavior is highly uneven: the median customer placed 3 orders, while some customers placed hundreds.
- Customer revenue is strongly right-skewed, with a small number of high-value customers contributing a large share of revenue.

- RFM scoring converts raw behavior into comparable 1–5 scores for Recency, Frequency and Monetary value.
- The six business-oriented segments reveal clear differences in customer engagement and historical value.
- Champions are the main revenue-driving group, while At Risk customers represent a historically valuable group that has become inactive.
- Potential Customers are recent but have low purchase frequency, making them a distinct early-repeat opportunity.
- Inactive / Low Value customers show both long recency and low historical frequency/value.

13. Important Limitations

- RFM segmentation is descriptive; it does not predict future purchases by itself.
- The segment rules were business-defined for this project and should be validated against actual campaign or retention results.
- Revenue is sales value, not profit or margin.
- Product-level findings describe historical purchasing behavior and do not prove that a product caused future purchases.
- The dataset contains wholesale activity, so large orders can strongly affect revenue-based metrics.